Traffic Accident Severity Prediction Using Machine Learning

Abstract

Traffic accidents remain a major public safety concern in large urban environments. Understanding the factors that influence accident severity can support better traffic management and safety policies. In this project, we investigate the use of machine learning techniques to predict accident severity using historical traffic accident data. Several classification models were implemented and evaluated, including Naive Bayes, Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting. In addition, ensemble approaches such as voting and stacking were explored to improve predictive performance. The models were evaluated using accuracy, weighted F1-score, precision, recall, and ROC-AUC metrics with cross-validation. The results show that ensemble models, particularly Gradient Boosting and stacking methods, achieve the strongest predictive performance. Feature importance analysis further highlights the influence of temporal factors, road infrastructure, and environmental conditions on accident severity.

1. Introduction

Road traffic accidents are a major global safety issue that leads to significant injuries, fatalities, and economic losses every year. In large urban areas such as Toronto, understanding the factors that contribute to accident severity is important for improving road safety and supporting data-driven decision making in transportation planning.

Traditional statistical methods have been used to analyze accident data, but these approaches may struggle to capture complex relationships between accident severity and the many environmental, temporal, and infrastructural factors involved. Machine learning techniques provide a powerful alternative, as they are capable of identifying nonlinear patterns and interactions within large datasets.

Recent research has shown that machine learning models such as Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting can effectively predict accident severity using historical traffic data. Ensemble learning methods in particular have demonstrated strong performance due to their ability to combine multiple models and reduce prediction errors.

In this project, we explore the application of several machine learning algorithms for accident severity prediction. The objective is to compare different classification models and determine which methods provide the most reliable predictions. By analyzing the performance of these models and examining feature importance, the project aims to identify key factors that influence accident severity.

2. Method

The project follows a standard machine learning workflow consisting of data preprocessing, feature engineering, model training, and model evaluation.

2.1 Data Preparation

The dataset includes multiple variables describing accident conditions, including environmental, temporal, and road-related features. These features include weather conditions, temperature, humidity, visibility, wind speed, and precipitation, as well as road infrastructure indicators such as traffic signals, crossings, junctions, and stop signs.

Before training the models, several preprocessing steps were applied. Missing values were handled where necessary, and categorical variables such as weather conditions and lighting conditions were transformed into numerical representations using one-hot encoding. This allowed the machine learning models to process categorical data effectively.

To simplify the classification task and focus on identifying severe accidents, the original severity levels were also grouped into two categories:

- **Low severity accidents**
- **High severity accidents**

This transformation enables the models to focus on distinguishing between minor incidents and more serious accidents.

2.2 Feature Engineering

Additional temporal features were extracted from the dataset to capture patterns related to time. These include:

- **hour of the day**
- **day of the week**
- **month of the year**
- **indicators for rush hour and nighttime conditions**

These features help capture behavioral and traffic patterns that may influence accident outcomes.

2.3 Machine Learning Models

Several classification models were implemented and compared:

- **Naive Bayes (baseline model)**
- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Gradient Boosting**

In addition to individual models, ensemble learning techniques were also explored:

- **Hard Voting**
- **Soft Voting**
- **Stacking (combining Random Forest and Gradient Boosting with Logistic Regression)**

Ensemble models combine the strengths of multiple algorithms and often provide more stable predictions.

3. Experiments

To evaluate model performance, the dataset was divided into training and testing sets. The models were trained using the training data and evaluated on unseen data.

Evaluation Metrics

Multiple evaluation metrics were used to measure model performance:

- **Accuracy**
- **Precision**
- **Recall**
- **Weighted F1-score**
- **ROC-AUC**

Using several evaluation metrics provides a more complete understanding of model performance, especially when dealing with classification tasks.

Cross-Validation

To ensure that the results are reliable and not dependent on a single data split, 5-fold cross-validation was applied to all models. This approach repeatedly trains and evaluates the models on different subsets of the data, providing a more robust estimate of model performance.

Hyperparameter Exploration

Several experiments were conducted to explore how model parameters influence performance. For example, the number of trees and tree depth were varied for the Random Forest model, while the learning rate and number of boosting iterations were adjusted for Gradient Boosting.

Error Analysis

An additional analysis was performed to examine prediction confidence and identify where the models make incorrect predictions. By examining false positives and false negatives, it is possible to understand which cases remain difficult for the models to classify correctly.

4. Results

The experimental results show clear differences in performance across the tested models.

The baseline model, Naive Bayes, achieved the lowest performance, with weighted F1-scores around 0.60. Logistic Regression performed slightly better, achieving approximately 0.65 F1-score. Decision Trees improved the results further, reaching approximately 0.69.

Tree-based ensemble methods provided significantly stronger performance. Random Forest achieved an F1-score of approximately 0.71, while Gradient Boosting achieved around 0.75. Gradient Boosting also achieved the highest ROC-AUC values, indicating strong classification capability.

The stacking model, which combines Random Forest and Gradient Boosting predictions using Logistic Regression, achieved the best overall performance. And here in the table is summary for all the results we have achieved.

Model	Accuracy	F1-W	Precision	Recall	ROC-AUC
Stacking (RF+GB→LR)	75.2%	0.752	0.753	0.752	0.843
Gradient Boosting	75.0%	0.750	0.752	0.750	0.834
Soft Voting	73.4%	0.733	0.741	0.734	0.819
Random Forest	71.7%	0.714	0.729	0.717	0.795
Hard Voting	73.4%	0.732	0.742	0.734	0.734
Decision Tree	69.4%	0.687	0.704	0.694	0.764
Logistic Regression	65.6%	0.656	0.657	0.656	0.710
Naive Bayes	61.2%	0.606	0.612	0.612	0.688

Table 4: All models — final test set results (sorted by ROC-AUC)

These results suggest that combining multiple models can improve predictive performance compared to individual models.

Feature importance analysis also revealed several influential predictors. The most important features included accident duration, distance, traffic signals, road crossings, time of day, and month of the year. These variables highlight the importance of both temporal patterns and road infrastructure in determining accident severity.

5. Conclusion

This project explored the use of machine learning models to predict traffic accident severity based on historical accident data. Several classification algorithms were implemented and compared, including both traditional models and ensemble learning methods.

The results demonstrate that ensemble methods such as Random Forest and Gradient Boosting outperform simpler models. Among all models tested, Gradient Boosting and the stacking ensemble achieved the best predictive performance.

Feature importance analysis indicates that accident duration, traffic infrastructure, and temporal factors are among the most influential predictors of accident severity.

Although the results are promising, several limitations remain. The dataset may contain missing information, and some relevant variables, such as driver behavior or vehicle characteristics, may not be included. Future work could explore more advanced models, incorporate additional data sources, or investigate deep learning approaches.

Overall, the findings suggest that machine learning techniques can provide valuable insights for traffic accident analysis and may support future efforts to improve road safety.

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